

Sequential Resampling under the Variable-Model Local Lemma: A Complete Witness-Tree Termination Proof

HCS-C325 theorem-and-evidence package

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Abstract

We give a convention-locked proof of the finite-variable Moser–Tardos theorem. Under the asymmetric witness criterion, every legal sequential violated-event rule terminates almost surely, avoids all bad events, and satisfies the exact per-event and total expectation bounds. The proof includes the resampling-table witness lemma and the complete branching sum. Exact finite chains audit, but do not prove, the general result.

1 Frozen variable model

Let $X_1, \dots, X_n$ be independent, finite-valued variables and let $\mathcal{A}$ be a finite family of bad events. Write $\text{vbl}(A)$ for the variables determining A , and $\Gamma(A) = \{B \neq A : \text{vbl}(A) \cap \text{vbl}(B) \neq \emptyset\}$. Each variable owns an independent table $X_i(0), X_i(1), \dots$ of copies. Initially the zeroth entries are exposed. At each step an arbitrary legal rule chooses a currently violated A and advances exactly the tables indexed by $\text{vbl}(A)$. The executable audit alone fixes lexicographic choice.

Theorem 1. *Suppose $x_A \in (0, 1)$ and*

$$\mathbb{P}(A) \leq x_A \prod_{B \in \Gamma(A)} (1 - x_B), \quad A \in \mathcal{A}. \quad (1)$$

Then every legal sequential rule terminates almost surely at an assignment avoiding $\mathcal{A}$. If N_A counts resamplings of A , then

$$\mathbb{E} N_A \leq \frac{x_A}{1 - x_A}, \quad \mathbb{E} \sum_A N_A \leq \sum_A \frac{x_A}{1 - x_A}.$$

The three exact binary examples contain seven events and all 112 assignment rows. Their rational absorbing-chain expectations and proper-tree series through size six are reconstructed independently. They are regression data, not an extrapolation to Theorem 1.

Revision certificate: frozen resampling dynamics and exact chains

This original round fixes the tables, dependencies, legal clock, theorem contract, and complete small-instance transition ledger.

References

- [1] R. A. Moser and G. Tardos, “A constructive proof of the general Lovasz Local Lemma,” *J. ACM* 57 (2010), Article 11; DOI: 10.1145/1667053.1667060; arXiv:0903.0544.